

TIME-LAPSE PHOTOGRAPHY

With Raspberry Pi 3

VIVEK G.R.

This project aims at developing a simple time-lapse photography device with Raspberry Pi 3 and Pi camera.

To get the most out of nature without tiring yourself, time-lapse photography is the best technique.

For time-lapse photography the film frames are captured at much lower rate than the normal speed used to view the frames in sequence. Here, a photographer takes several pictures while the camera is locked in a still position for a

preset duration that can vary from half an hour to several days. When it is played at normal speed, time appears to be moving faster. One example where time-lapse photography can be used is the scene of the Sun rising (Fig. 1) or setting.

How it works

The process is simple. A camera fitted on a tripod is placed from where the natural scenery can be shot. Then, the duration is fixed and the camera is left to do the job.

After the camera has finished taking photographs for the preset duration, you can download all images into your PC and analyse the flow of photos with respect to time. Then, there are two options:

1. You select individual photos among the bunch of photos taken by time-lapse technique and send these for further processing.
2. You combine all photos to make it into a video (Fig. 2), depicting how nature changes with respect to time.

Applications of time-lapse photography include:

- Landscapes and celestial motion
- Plants growing and flowers blooming
- Fruits rotting
- Evolution of a construction project
- People in a city

The following are required to get started with time-lapse photography:

1. Raspberry Pi 3
2. SD card with latest Raspbian OS installed
3. Pi camera
4. Power adaptor (5V, 2.5A)
5. Mouse
6. Keyboard
7. HDMI to VGA adaptor
8. Standard PC monitor

Setup

To set up please follow the steps below:

Enable Pi camera. Connect Pi camera with Raspberry Pi 3 (Fig. 3). Connect the monitor to Raspberry Pi via VGA to HDMI converter and also plug in the USB mouse and keyboard.

Boot up Raspberry Pi 3. You will see Pi desktop (assuming you have already installed Raspbian OS). Open the terminal and type the following:

```
$ sudo raspi-config
```

Select Enable Camera, Finish and Yes to reboot (Fig. 4).

Install Python support.

To install a Python support for Pi camera, type the following command:

```
$ sudo apt-get install python-pi-camera python3-pi-camera
```

```
$ sudo apt-get install ffmpeg
```

Start Python environment by entering the following:

```
$ sudo idle &
```

This will bring up Python environment. Python is an easy language to learn and



Fig. 1: Time-lapse photography of the Sun rising

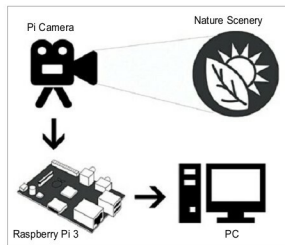


Fig. 2: Time-lapse architecture



Fig. 3: Pi camera setup with Raspberry Pi 3